

RAYMOND R WAYESU

S T A T I S T I C I A N



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Kampala, Uganda

PROFILE

Raymond is a statistician, biostatistician and data scientist with more than 10 years of experience that include coding skills in Python and R, randomized control trials (Phase 1 and Phase 2) and experimental design, machine learning and data analysis.

SKILLS

- R
- Python
- SQL (Databases)
- Linux
- Microsoft Office

EXPERIENCE

BIOSTATISTICIAN

Uganda Virus Research Institute -
project of Coalition for Epidemic Preparedness Innovations
2024 July - Present

At the Uganda Virus Research Institute (UVRI), a leading East African Center of Excellence in Virology, I serve as a Biostatistician applying advanced statistical methods to immunology studies while working closely with investigators on research design and data interpretation. UVRI focuses on viral infections including Ebola, Marburg and Mpox, conducting crucial research that informs global response strategies. Our work generates vital genetic and immunological data that shapes control measures worldwide, positioning UVRI at the forefront of regional and international public health efforts.

As a Biostatistician, I apply advanced statistical methodologies to ongoing Immunology studies. I collaborate closely with project leads and Principal Investigators, to shape the design, analysis, and interpretation of data, playing a crucial role in optimising assays for detecting responses to Ebola, Marburg, and other emerging threats, thus enhancing preparedness for epidemics and future vaccine trials.

EDUCATION

BACHELOR OF STATISTICS

Makerere University Kampala
2008 - 2011

MASTER OF SCIENCE STATISTICS

Linkoping University Sweden
2018 - 2020

GOOD CLINICAL PRACTICE

NIDA Clinicals Trials Network
2023

PUBLICATIONS

ORCID

<https://orcid.org/0009-0007-7797-4339>

"Forecasting anomalies in time series data from online production environments"

<http://urn.kb.se/resolve?urn=urn:nbn:se:liu:diva-166044>

Two Decades Later: Long-Term Multisystem Sequelae and Subclinical Organ Dysfunction in Sudan Ebola Virus (SUDV) Survivors. Persistent Sequelae in Sudv Survivors after 25 Years

<https://ssrn.com/abstract=5259807>

Resilience and Residuals Beyond Containment: The Hidden Burden of Bundibugyo Ebola Virus Survivorship Sixteen Years On.

MEMBERSHIPS

American Statistical Association

ASA ID: 170972

EXPERIENCE

STATISTICIAN

Elgon Centre for Health Research and Innovation - Department of Anaesthesia & Critical Care

2024 September - Present

Elgon Centre for Health Research and Innovation, part of the Elgon International Health Institute (EIHI), is dedicated to addressing health challenges through research, innovation, and training of health professionals. EIHI is particularly known for its focus on middle-level multi-skilled health cadres essential for improving healthcare delivery in Uganda.

I implemented statistical modelling, machine learning and interactive dashboards.

ASSISTANT BIOSTATISTICIAN

Makerere University - iTECH Project

2024 February - 2024 June

Previously, as Assistant Biostatistician for the iTECH project at Makerere University Hospital, I ensured data quality and compliance with research protocols while conducting statistical analyses and preparing study reports. This project develops innovative clinical tools for high-risk pregnancy management, aiming to reduce stillbirth rates across Uganda, sub-Saharan Africa, and beyond. The research examines maternal markers, cardiovascular function, and ultrasound data to understand stillbirth pathophysiology in resource-limited settings.

I wrote over 100 R scripts and Python scripts to analyze and interpret data to identify trends, patterns, and potential areas for improvement applicable to iTECH prediction modelling for still births.

EXPERIENCE

HUB DATA SUPERVISOR

Sanyu Africa Research Institute

2023 September - 2024 January

The Babygel Study is a cluster-controlled trial examining the impact of improving hand hygiene in rural Ugandan households using locally- produced alcohol-based hand gel. It is recruiting 6000 participants from hard-to-reach communities in 72 villages in the Mbale and Budaka Districts of Eastern Uganda. In these villages, women are being given a birth Kit and half are being given BabyGel after delivery. The gel is made locally in Uganda from sugar cane and, if the trial proves successful, it could be included in delivery packs for every expectant mother. The study's primary outcome measure is severe infant illness or death in the first 90 days of life.

The main role of the hub data supervisor is to coordinate the hub data collectors, ensure that quality data is collected by hub data collectors, quality data is entered in to the electronic data capture system.

STATISTICIAN

ICS Engineering and Environment

2021 November - 2022 December

- Worked as a Statistician for the East African Crude Oil Pipeline (EACOP) RAPI project - Uganda section
- Advised the data manager on procedures to cleaning, validation and maintenance of high-quality data
- Performed descriptive and exploratory statistical analysis of individual-level or aggregate-level data
- Performed statistical analysis on Socio-economic, Livelihood and vulnerability data as required by Project Manager and Data Manager.
- Created statistical models that benefited the decision making within the Project

DATA SCIENCE DEVELOPMENT INTERN

Infor (Sweden) AB



2020 January - 2020 June

- Analyzed big data collected from North America and Europe
- Worked with dashboards
- Presented statistical results on a weekly basis
- Used R and Python to analyze and interpret data to derive useful information from the data and developed inferences and conclusions that were published

Publication:

<http://urn.kb.se/resolve?urn=urn:nbn:se:liu:diva-166044>

EXPERIENCE

PROJECT SPECIALIST - STATISTICS

One Acre Fund - Uganda

2017 August - 2018 May

- Conducted Phase 1, Phase 2, Phase 3 Randomized Control Trials for agricultural control and treatments
- Collaborated with principal investigators (research and development leads) in developing study methodology, sampling strategies, data collection and management and statistical approaches for example in sample size calculations and effective size calculations and testing for "generalizability"
- Used R and wrote many R scripts that included t-tests, Tukeys HSD, ANOVA, F-test, non-parametric tests and used many R packages
- Responsible for developing the data management guideline and advised the department lead on procedures to oversee the collection, cleaning, validation and maintenance of high-quality research data.
- Conducted trainings in basic statistics for less experienced employees that I supervised in my department engaged in the collection, tabulation and analysis of statistical data.

BUSINESS DATA ANALYST

Baridi Stores Limited (currently part of Haba Na Haba Food Bank)

2014 September - 2017 April

- Led to the business winning 3 awards 3 years in a row, 2014, 2015 and 2016
- Created a detailed business analysis, outlining problems, opportunities and solutions for the business
- Planning and monitoring
- Financial modelling using statistical methods
- Defined business requirements and reporting them back to stakeholders

DATA INTEGRITY OFFICER

National Social Security Fund

2012 April - 2014 September

- The problem was ensuring improved financial transparency for pension savings.
- Cleaned, analyzed and manipulated data, provided action points regarding large sets of data related to pension in line with specified metrics.